Block practice · Integration, areas, volumes and kinematics

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Finding a curve from its gradient function

Velocity and displacement by integrating

Distance travelled by integrating velocity

Areas of regions made of several pieces

OLevels.com SCHOOL

Practice workbook · Finding a curve from its gradient function

Integrating dy/dx term by term, then using given points (and sometimes the factor theorem) to find the constant and any unknowns.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-16

1. Question 7 · part a.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a.i only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

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20

1. Question 7 · part a.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part a · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

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2. Question 9 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part whole · 9 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

8 $f'(x) = 18x^2 - 2x + 13$

Given that $(2x - 1)$ is a factor of $f(x)$

show that the curve with equation $y = f(x)$ has only one intersection with the x-axis.

(9)

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3. Question 8 · part whole · 9 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

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Question 8 continued

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(Total for Question 8 is 9 marks)

3. Question 8 · part whole · 9 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part a · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

10 The equation of a curve C is $y = f(x)$ where $f'(x) = 3x^2 - 4x - p$ and $p \neq 0$

The points with coordinates $(2, 0)$ and $(-1, 9)$ lie on C.

(a) Show that C has equation $y = x^3 - 2x^2 - 4x + 8$

(6)

The straight line l has equation $y = 8 - 4x$

(b) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(6)

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4. Question 10 · part a · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part a · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

8

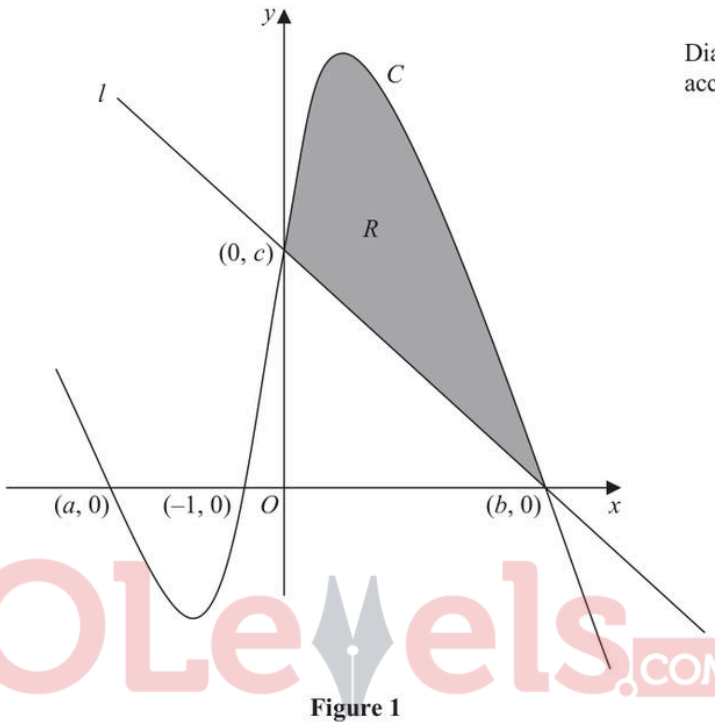


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation $y = f(x)$

The curve C passes through the points with coordinates

$(a, 0)$, $(-1, 0)$, $(b, 0)$ and $(0, c)$

Given that $f'(x) = 17 + 2x - 3x^2$

(a) show that the equation of C is $y = 15 + 17x + x^2 - x^3$

(4)

(b) Find the value of a, the value of b and the value of c

(6)

The straight line l intersects C at the points with coordinates $(b, 0)$ and $(0, c)$

The region R, shown shaded in Figure 1, is bounded by l and C

(c) Use algebraic integration to find the exact area of region R

(5)

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OLevels.COM SCHOOL

P 7 1 8 1 8 A 0 1 8 2 8

5. Question 8 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part a · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

3 $g'(x) = mx^2 - 10x - 37$ where m is an integer

The curve $y = g(x)$ passes through the point with coordinates $(1, 20)$

Given that $(x - 5)$ is a factor of $g(x)$

(a) show that $g(x) = 2x^3 - 5x^2 - 37x + 60$

(5)

(b) Hence, or otherwise, use algebra to solve the equation $g(x) = 0$

(3)

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6. Question 3 · part a · 5 marks · Working and answer

Working and answer

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Practice workbook · Velocity and displacement by integrating

Integrating acceleration to get velocity, and velocity to get displacement, using starting conditions for the constants.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-28

1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.

5 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3t^2 - 23t + 30$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

When $t = 0$, P is at the point with coordinates $(d, 0)$

Given that, when $t = 8$, P is at the point with coordinates $(26, 0)$

(c) find the value of d

(4)

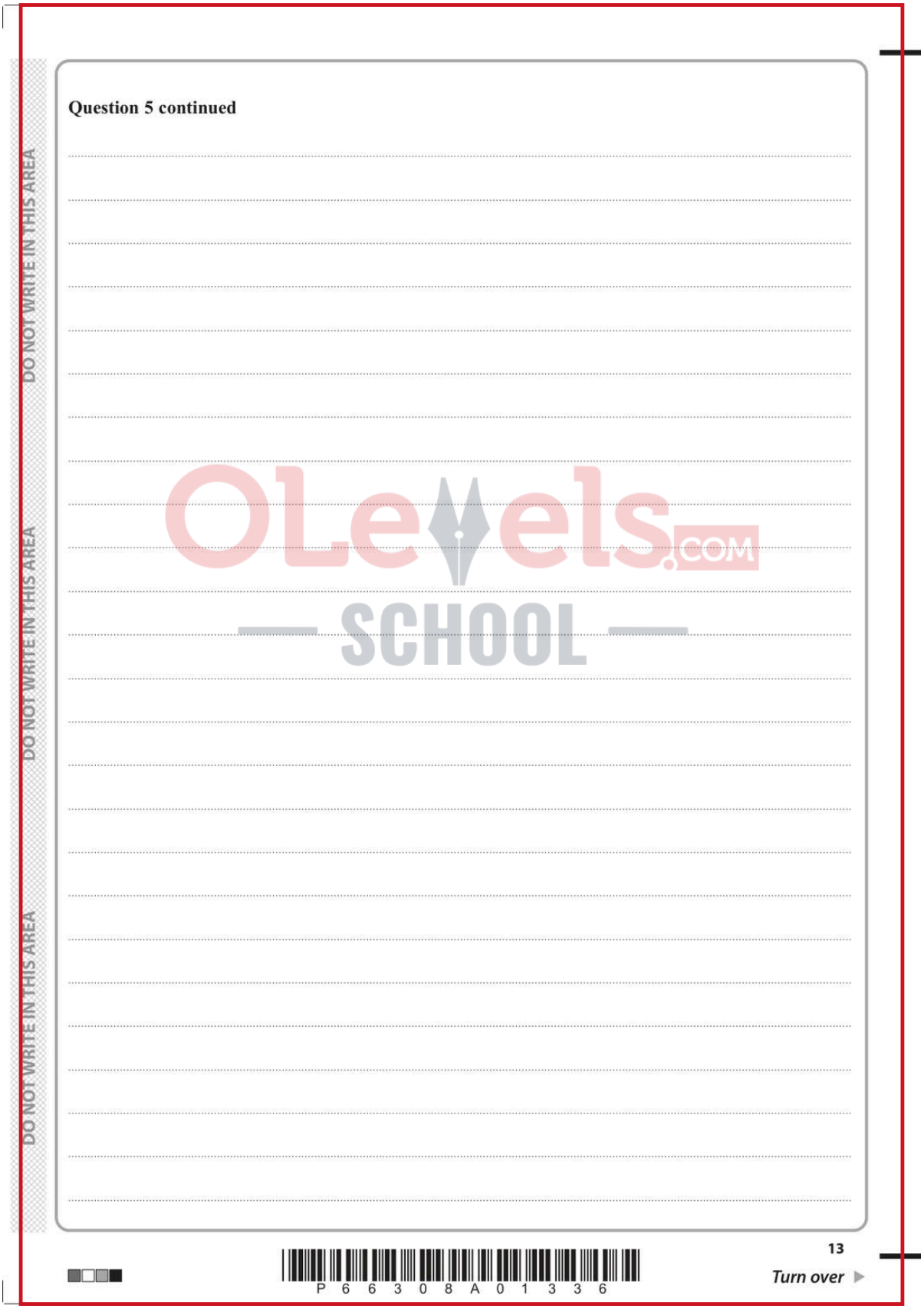
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1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.



1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 9 marks)

1. Question 5 · part c · 4 marks · Working and answer

Working and answer

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2. Question 3 · part a.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part a.ii only.

3 A particle P moves along the x-axis.

At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 6t - 16$

When $t = 0$, P is at the origin and is moving with velocity 12 m/s.

(a) Find an expression in terms of t for

(i) the velocity of P at time t seconds

(ii) the displacement of P at time t seconds.

(4)

(b) Hence find the time at which P first returns to the origin.

(3)

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P 7 3 5 8 6 A 0 6 3 2

2. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 A particle P moves along the x-axis. At time t seconds ($t \geqslant 0$) the acceleration, a m/s^2, of P is given by $a = 6t - 12$

When $t = 0$, P is at rest at the origin.

(a) Find the velocity of P when $t = 2$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T.

(2)

(c) Find the distance travelled by P in the first 8 seconds of its motion.

(3)

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3. Question 4 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 5 · part a · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a only.

5 A particle P is moving along the x-axis.
At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 3t - 4$

When $t = 0$, P is at rest.

(a) Find the velocity of P when $t = 4$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T

(2)

When $t = 0$, P is at the point with coordinates $(-10, 0)$

(c) Find the displacement of P from the origin when $t = 3$

(4)

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4. Question 5 · part a · 3 marks · Working and answer

Working and answer

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5. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

5 A particle P is moving along the x-axis.
At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 3t - 4$

When $t = 0$, P is at rest.

(a) Find the velocity of P when $t = 4$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T

(2)

When $t = 0$, P is at the point with coordinates $(-10, 0)$

(c) Find the displacement of P from the origin when $t = 3$

(4)

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5. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

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Question 5 continued

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(Total for Question 5 is 9 marks)

5. Question 5 · part c · 4 marks · Working and answer

Working and answer

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6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

8 A particle P moves along the positive x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3 + 5t - 2t^2$

At time t seconds, P is at the point with coordinates $(x, 0)$.

Given that at time $t = 0$, P is at the point with coordinates $(5, 0)$, find the maximum value of x, justifying that this is a maximum value.

(8)

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P 6 0 4 7 5 A 0 2 0 3 6

6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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22

P 6 0 4 7 5 A 0 2 2 3 6

6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 8 marks)

6. Question 8 · part whole · 8 marks · Working and answer

Working and answer

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7. Question 2 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 A particle P is moving along a straight line, which passes through the fixed point O.

At time t seconds ($t \geqslant 0$), the velocity, v m/s, of P is given by

$$v = t^2 - 3t + 4$$

At time t seconds the acceleration of P is a m/s^2

(a) Find an expression for a in terms of t

(2)

The displacement of P from O is 7 m when $t = 2$

(b) Find the exact displacement of P from O when $t = 4$

(5)

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7. Question 2 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.



Question 2 continued

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(Total for Question 2 is 7 marks)

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7. Question 2 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 7 · part b · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

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8. Question 7 · part b · 5 marks · Working and answer

Working and answer

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Practice workbook · Distance travelled by integrating velocity

Finding total distance by integrating velocity, splitting at times of rest and adding sizes.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-29

1. Question 3 · part b · 4 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part b only.

3 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = t^2 - 4t + 7$$

(a) Find the acceleration of P, in m/s^2, when $t = 3$

(2)

(b) Find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 6$

(4)

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1. Question 3 · part b · 4 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part b only.



Question 3 continued

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(Total for Question 3 is 6 marks)

1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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2. Question 1 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

1 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = 3t^2 - 9t + 7$$

(a) Show that P never comes to rest.

(2)

(b) Find the acceleration of P, in m/s^2, when $t = 4$

(2)

Using algebra

(c) find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 5$

(3)

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P 7 2 8 6 5 A 0 4 3 6

2. Question 1 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

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Question 1 continued

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(Total for Question 1 is 7 marks)

2. Question 1 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

4 A particle P moves along the x-axis. At time t seconds ($t \geqslant 0$) the acceleration, a m/s^2, of P is given by $a = 6t - 12$

When $t = 0$, P is at rest at the origin.

(a) Find the velocity of P when $t = 2$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T.

(2)

(c) Find the distance travelled by P in the first 8 seconds of its motion.

(3)

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3. Question 4 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 8 marks)

3. Question 4 · part c · 3 marks · Working and answer

Working and answer

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4. Question 4 · part b · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

4 A particle P is moving along the x-axis. At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by $v = 2t^2 - 16t + 30$

(a) Find the acceleration, in m/s^2, of P when $t = 5$

(2)

P comes to instantaneous rest at the points M and N at times t_1 seconds and t_2 seconds where $t_2 > t_1$

(b) Find the exact distance MN

(8)

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4. Question 4 · part b · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 10 marks)

4. Question 4 · part b · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

4 A particle P is moving along a straight line. At time t seconds $(t \geqslant 0)$, its velocity, v m/s, is given by $v = 4t^2 - 6t + 5$

(a) Find the minimum speed of P

(3)

The acceleration of P at time T seconds is $18\ \text{m/s}^2$

(b) Find the value of T

(2)

When $t = 0$, P is at the point X, and when $t = 3$, P is at the point Y

(c) Find the distance XY

(4)

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P 7 8 9 6 5 A 0 8 3 2

5. Question 4 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

5. Question 4 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part c · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 A particle P is moving along the x-axis.

At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

(a) Find the acceleration of P when $t = 5$

(2)

The particle comes to instantaneous rest at the points A and B at times t_1 seconds and t_2 seconds respectively, where $t_1 < t_2$

(b) Find the value of t_1 and the value of t_2

(2)

(c) Use calculus to find the distance AB

(3)

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6. Question 5 · part c · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 7 marks)

6. Question 5 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 5 · part c · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

5 A particle P is moving along a straight line.

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 3t^2 - 16t + 5$$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

(c) Find the distance, in m, that P travels in the interval $1 \leqslant t \leqslant 4$

(5)

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7. Question 5 · part c · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

7. Question 5 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part c · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

3 A particle P is moving along a straight line, from a fixed origin O

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

At time t seconds the acceleration of P is a m/s^2

(a) Find the value of t for which $a = 0$

(3)

P is instantaneously at rest at time T_1 seconds and at time T_2 seconds where $T_2 > T_1$

(b) Find the value of T_1 and the value of T_2

(3)

(c) Find the exact distance, in metres, that P travels between the times T_1 and T_2
Show your working clearly.

(4)

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8. Question 3 · part c · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 10 marks)

8. Question 3 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Areas of regions made of several pieces

Finding the area of a region built from an integral plus or minus a triangle, or from two curves each on its own interval.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-21

1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

11

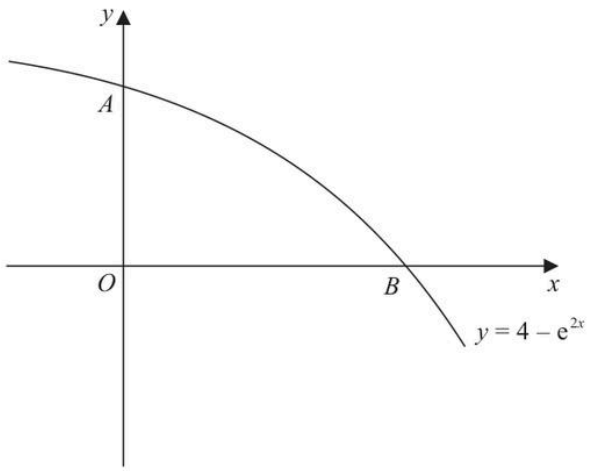


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 11 continued

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1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 11 continued

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1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 11 continued

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1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 14 marks)

1. Question 11 · part c · 7 marks · Working and answer

Working and answer

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2. Question 11 · part b · 11 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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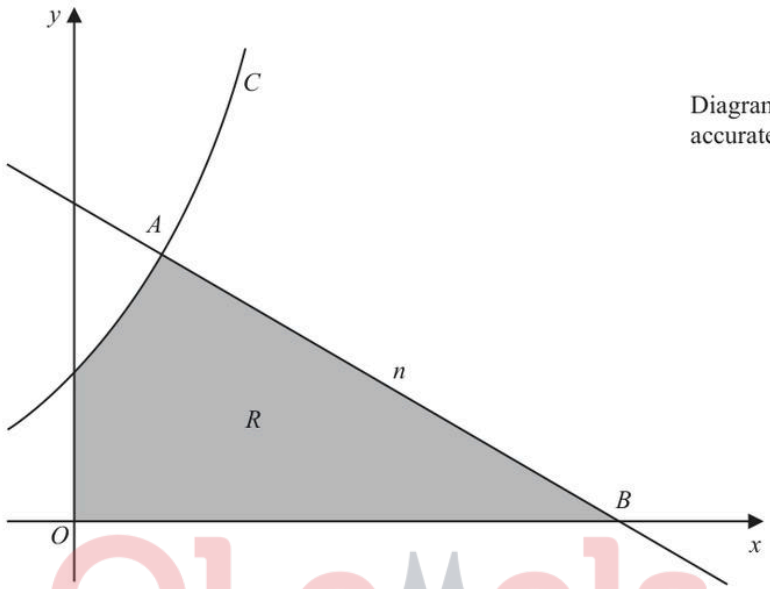


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a curve C with equation $y = 3e^{\frac{x}{3}}$
The point A with coordinates $(2, p)$ lies on C

(a) Write down the exact value of p

(1)

The straight line n, shown on Figure 4, is the normal to C at A and crosses the x-axis at point B

The region R, shown shaded in Figure 4, is bounded by C, n, the x-axis and the y-axis.

(b) Use algebraic integration to find the area of R

Give your answer in the form $\frac{W}{V}e^2 + We^{\frac{2}{3}} - W$

where W and V are integers to be found.

(11)

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2. Question 11 · part b · 11 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 11 continued

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2. Question 11 · part b · 11 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 12 marks)

2. Question 11 · part b · 11 marks · Working and answer

Working and answer

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3. Question 7 · part c · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

7

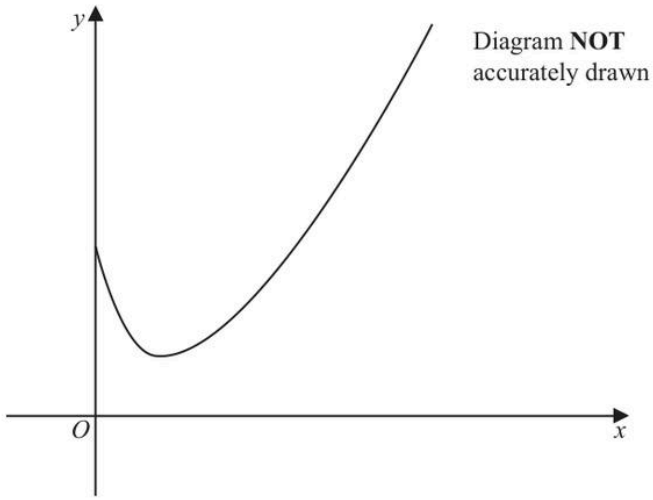


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{x^2}{4} - 3\sqrt{x} + 8$$

The point P lies on C and has coordinates $(4, a)$

(a) Show that $a = 6$

(1)

The line L is the normal to C at the point P

(b) Show that an equation of L is $5y + 4x - 46 = 0$

(6)

The finite region R is bounded by the curve C, the line L, the x-axis and the line with equation $x = 1$

(c) Use calculus to find the exact area of R

(6)

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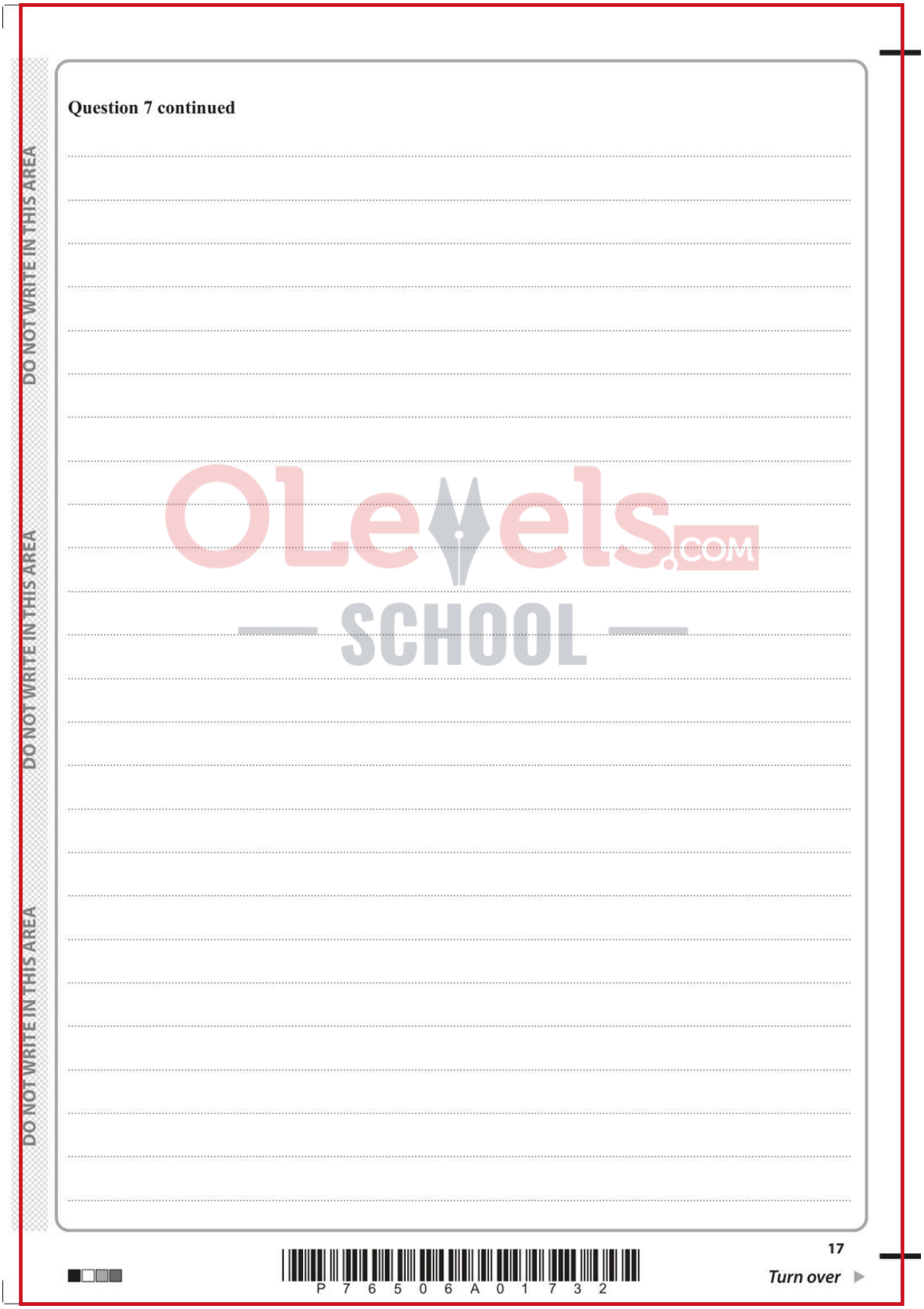
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3. Question 7 · part c · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.



Question 7 continued

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Turn over

3. Question 7 · part c · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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3. Question 7 · part c · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 13 marks)

3. Question 7 · part c · 6 marks · Working and answer

Working and answer

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4. Question 5 · part b · 7 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

5

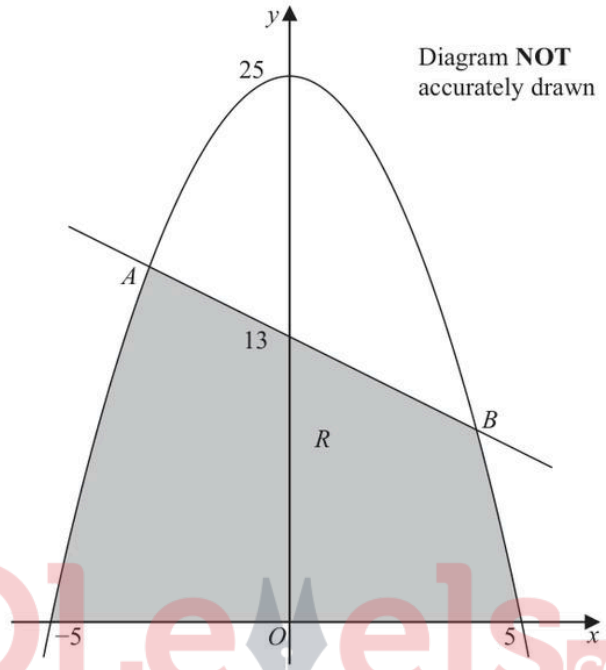


Figure 2

Figure 2 shows part of the curve with equation $y = 25 - x^2$ and part of the line with equation $y + x = 13$

The curve and the line intersect at the points A and B.

(a) Use algebra to find the coordinates of A and the coordinates of B.

(4)

The region R, shown shaded in Figure 2, is bounded by the curve, the straight line and the x-axis.

(b) Use algebraic integration to find the area of R.

(7)

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4. Question 5 · part b · 7 marks

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Further practice · Attempt part b only.

Question 5 continued

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4. Question 5 · part b · 7 marks

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Further practice · Attempt part b only.

Question 5 continued

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4. Question 5 · part b · 7 marks

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Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 11 marks)

4. Question 5 · part b · 7 marks · Working and answer

Working and answer

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5. Question 9 · part c · 8 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

9 (a) Using a formula given on page 2, show that

$$\cos 2\theta = 2\cos^2\theta - 1$$

(2)

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \left(2\cos^2\theta - 1\right) \mathrm{d}\theta = -\frac{a+\sqrt{b}}{c}$$

where a, b and c are integers to be found.

(4)

Diagram **NOT** accurately drawn

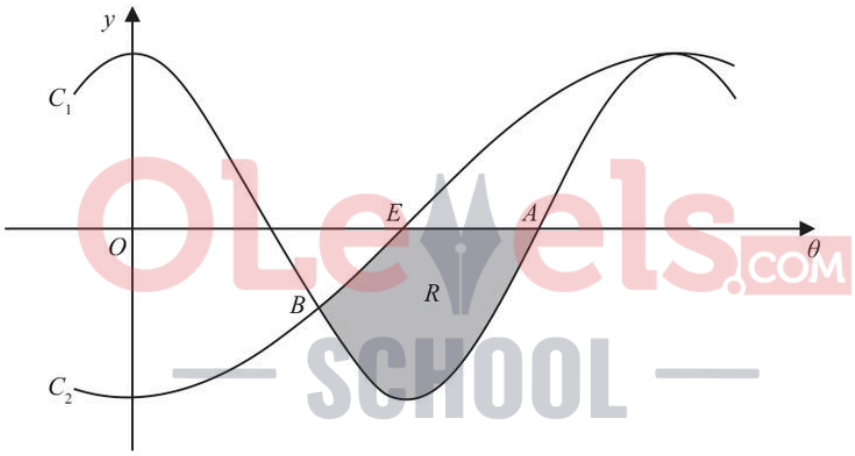


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 2\cos^2\theta - 1$ and part of the curve C_2 with equation $y = -\cos\theta$

Point B is the intersection of C_1 and C_2 as shown in Figure 3

Point $A\left(\frac{3\pi}{4}, 0\right)$ is the intersection of C_1 with the θ-axis as shown in Figure 3

Point $E\left(\frac{\pi}{2}, 0\right)$ is the intersection of C_2 with the θ-axis as shown in Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the θ-axis, C_1 and C_2

(c) Use calculus to find, in its simplest form, the exact area of R

(8)

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5. Question 9 · part c · 8 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 8 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 8 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 14 marks)

5. Question 9 · part c · 8 marks · Working and answer

Working and answer

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6. Question 12 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

12

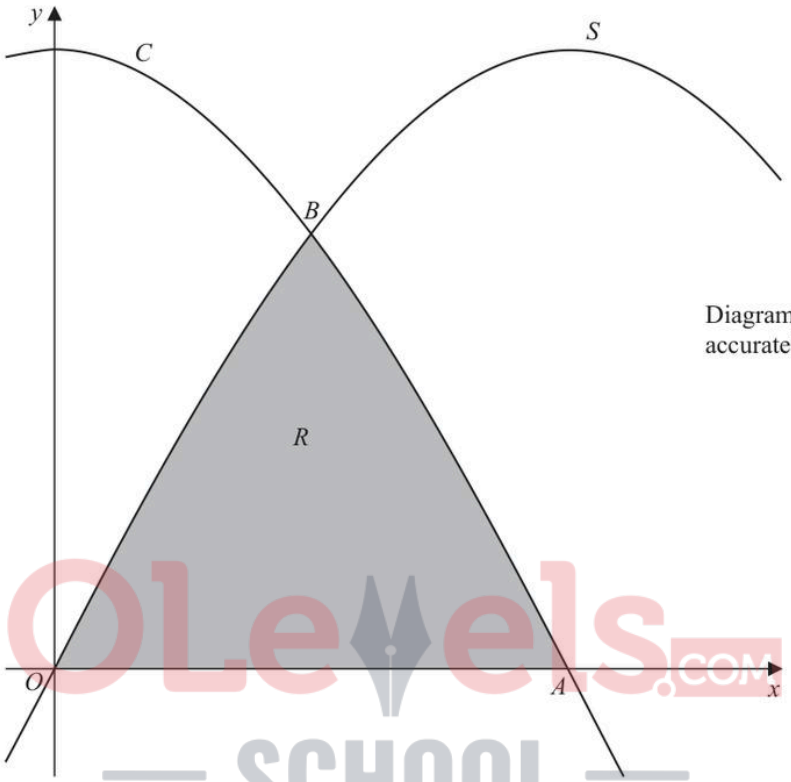


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the x-axis, the curve S with equation $y = 2\sin x$ and the curve C with equation $y = 2\cos x$.
As shown in Figure 2, C crosses the x-axis at the point A.

(a) Write down the x coordinate of A.

(1)

As shown in Figure 2, C and S intersect at the point B.

(b) Find the x coordinate of B.

(2)

(c) Using calculus, find the area of the shaded region R.
Give your answer in the form $a - \sqrt{b}$ where a and b are integers.

(4)

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6. Question 12 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 12 continued

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6. Question 12 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 12 continued

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(Total for Question 12 is 7 marks)

6. Question 12 · part c · 4 marks · Working and answer

Working and answer

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